KERALA POLYTECHNIC · REVISION 2026

Machine Learning and Neural Network

Complete student lesson for course code **4342**.

Revision 2026 Semester 4 Program Core 4 modules

Course snapshot

Scheme: 4 (L:4,T:0,P:0); 60 periods

Credits: 4

Contact: 4 (L:4,T:0,P:0)

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.
2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

4342

Semester

4

Credits

4

Teaching scheme

4 (L:4,T:0,P:0); 60 periods

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Supervised machine learning techniques	16	CO1 · Applying
2	Clustering and dimensionality reduction	14	CO2 · Applying
3	Evaluation metrics for machine-learning models	12	CO3 · Applying
4	Artificial Neural Networks	16	CO4 · Applying

Prerequisites

- Engineering Mathematics I & II; Artificial Intelligence.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. Students will be able to get basic knowledge on the techniques to build an intellectual machine and apply machine learning algorithms to solve problems. This course also summarizes the basic concepts of artificial neural networks.

Course Outcomes

1. Demonstrate supervised machine learning techniques.
2. Demonstrate clustering and dimensionality reduction techniques.
3. Demonstrate evaluation metrics for machine learning models.
4. Demonstrate the concept of Artificial Neural Networks.

CO–PO mapping as supplied

CO–PO mapping is reproduced from the supplied syllabus; the numerical mapping values are retained exactly where stated.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Machine learning concepts	1
module-1	M1.02	Process involved in machine learning	2
module-1	M1.03	Supervised learning	2
module-1	M1.04	Classification techniques	4
module-1	M1.05	Regression techniques	4
module-1	M1.06	Data preprocessing	2
module-2	M2.01	Types of clustering	2
module-2	M2.02	Clustering techniques	5
module-2	M2.03	Dimensionality reduction techniques	7
module-3	M3.01	Reinforcement learning	4
module-3	M3.02	Q-learning	2
module-3	M3.03	Regression evaluation metrics	2
module-3	M3.04	Classification evaluation metrics	2
module-3	M3.05	Clustering evaluation and validation	2
module-4	M4.01	Artificial neural-network concepts	3
module-4	M4.02	Perceptron and multilayer perceptron	3
module-4	M4.03	Back-propagation algorithm	3
module-4	M4.04	Recurrent networks	3
module-4	M4.05	Loss and activation functions	2
module-4	M4.06	Batch normalization	2

Learning Roadmap

1. Supervised machine learning techniques

2. Clustering and dimensionality reduction

3. Evaluation metrics for machine-learning models

CO1 · Applying · 16 hours

CO2 · Applying · 14 hours

CO3 · Applying · 12 hours

4. Artificial Neural Networks
CO4 · Applying · 16 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Supervised machine learning techniques

Machine learning builds a model from data so that it can make useful predictions or decisions. The official module covers the machine-learning process, learning categories, supervised methods, classification, regression and preprocessing.

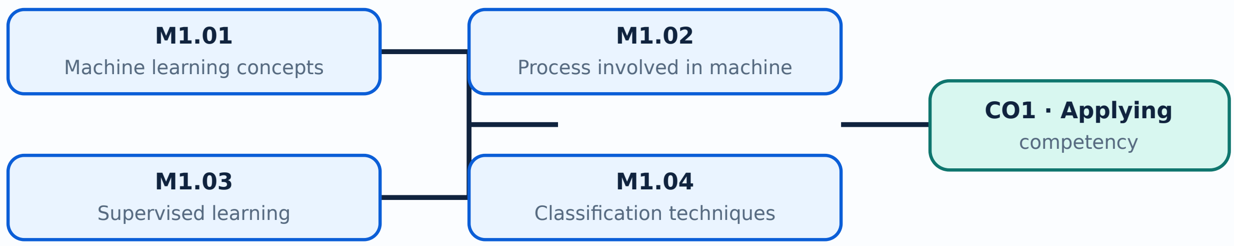
Hours: 16
CO1 · Applying · Bloom level: Applying

Learning goals

- Explain machine-learning concepts and applications.
- Select a supervised method for a stated problem.
- Prepare data before training and testing a model.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

M1.01 — Machine learning concepts (1 hours)

Concept and explanation

Machine learning is the study of algorithms that learn patterns from examples rather than being programmed with every rule. Applications include prediction, classification, recommendation, anomaly detection and automation.

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Study points

- Define machine learning.
- State practical applications.
- Distinguish a model from the data used to train it.

> **Engineering / workplace application:** Quality inspection, demand forecasting and document classification use learned models.

M1.02 — Process involved in machine learning (2 hours)

Concept and explanation

A typical process includes defining the problem, collecting and understanding data, cleaning and preprocessing, selecting features, dividing data into training and testing sets, training a model, evaluating it and monitoring performance.

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Study points

- Arrange the major steps in order.
- Explain why validation is needed.
- Separate training information from test information.

Suggested procedure

1. Define the target and success measure.
2. Inspect and preprocess the dataset.
3. Train, validate, evaluate and document the model.

M1.03 — Supervised learning (2 hours)

Concept and explanation

Supervised learning uses labelled examples containing input features and a known target. Classification predicts categories, whereas regression predicts a numerical value.

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Study points

- Identify features and labels.
- Distinguish classification from regression.
- Give one example of each.

> **Engineering / workplace application:** A labelled set of machine readings can be used to predict a fault class or remaining life.

M1.04 — Classification techniques (4 hours)

Concept and explanation

Classification methods include logistic regression, K-nearest neighbours, support vector machines, Naive Bayes, decision trees and random forests. Model choice depends on data type, interpretability, scale, noise and performance requirements.

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Study points

- Explain the purpose of a classifier.
- Compare at least two classification methods.
- Relate a decision boundary to a prediction.

> **Illustrative example:** For a two-class problem, a classifier assigns a new feature vector to one of the learned classes; its decision can then be checked with a confusion matrix.

M1.05 — Regression techniques (4 hours)

Concept and explanation

Linear regression models a numerical response using one or more input variables. Multiple linear regression extends the model to several predictors and requires attention to scale, correlation, residuals and overfitting.

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Study points

- State the idea of linear regression.
- Distinguish simple and multiple regression.
- Interpret a prediction and its error.

Engineering / workplace application: Energy consumption and production yield can be estimated from operating variables.

M1.06 — Data preprocessing (2 hours)

Concept and explanation

Preprocessing converts raw data into a form suitable for learning. Normalization brings variables to comparable ranges, while missing-value handling may use deletion, imputation or a domain-informed replacement.

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Study points

- Explain normalization.
- Choose a cautious missing-value strategy.
- Avoid data leakage during preprocessing.

Common mistake: Do not calculate preprocessing parameters from the test set and then report an unrealistically high test performance.

Module summary: A reliable workflow moves from problem definition and data preparation to training, validation, evaluation and deployment.

OFFICIAL MODULE 2

Clustering and dimensionality reduction

Unsupervised methods discover structure without a supplied target label. The module covers clustering families, K-means and mean-shift, followed by dimensionality-reduction methods.

Hours: 14
CO2 · Applying · Bloom level: Applying

Learning goals

- Classify common clustering approaches.
- Apply the main steps of a clustering method.
- Explain why dimensionality reduction is useful.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.

Learning map for Module module-2: the listed topics build the module competency.

M2.01 — Types of clustering (2 hours)

Concept and explanation

Clustering groups observations so that members of a group are more similar to one another than to members of other groups. Hierarchical, agglomerative, divisive and partitional approaches differ in how groups are formed and represented.

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Study points

- Define a cluster.
- Distinguish hierarchical and partitional clustering.
- Read a dendrogram conceptually.

M2.02 — Clustering techniques (5 hours)

Concept and explanation

K-means iteratively assigns points to the nearest centroid and updates centroids. Mean-shift moves candidate centres toward regions of high density. Initialisation, distance measure, scaling and the number of clusters affect the result.

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Study points

- State the K-means sequence.
- Explain the role of a centroid.
- Compare K-means and mean-shift.

Suggested procedure

1. Scale or normalize suitable features.
2. Choose initial parameters.
3. Assign points, update centres and repeat until stable.
4. Inspect the clusters for engineering meaning.

M2.03 — Dimensionality reduction techniques (7 hours)

Concept and explanation

Principal Component Analysis creates orthogonal directions of maximum variance. Factor analysis models observed variables through latent factors, multidimensional scaling preserves pairwise distances, and Linear Discriminant Analysis seeks class-separating directions when labels are available.

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Study points

- Explain the purpose of reducing dimensions.
- Compare PCA with LDA.
- Interpret a reduced representation cautiously.

Engineering / workplace application: Visualization, noise reduction and faster model training are common uses of dimensionality reduction.

Module summary: Unsupervised analysis is exploratory: the result must be interpreted against the data and engineering context.

OFFICIAL MODULE 3

Evaluation metrics for machine-learning models

Evaluation connects a model’s numerical output to a meaningful performance statement. The official content includes reinforcement learning, Q-learning and metrics for regression, classification and clustering.

Hours: 12
CO3 · Applying · Bloom level: Applying

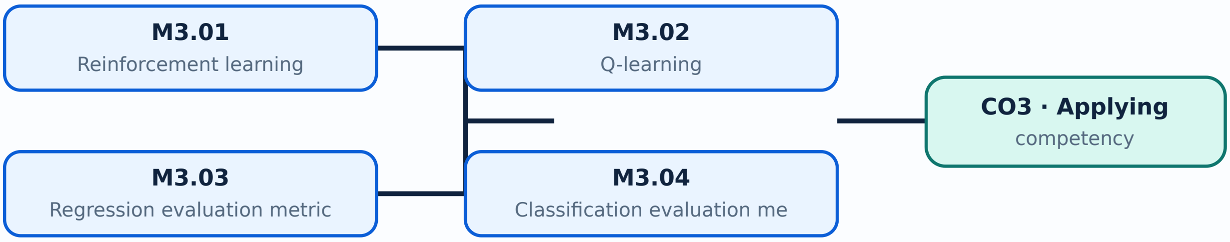
Learning goals

- Select a metric that matches the task.

- Interpret errors and confusion-matrix measures.
- Use validation and resampling to estimate generalisation.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

M3.01 — Reinforcement learning (4 hours)

Concept and explanation

Reinforcement learning involves an agent, environment, state, action, reward and policy. The agent learns through interaction rather than a fixed labelled dataset; the objective balances exploration and exploitation.

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Study points

- Define the key elements.
- Distinguish reinforcement learning from supervised learning.
- Explain exploration and exploitation.

Engineering / workplace application: A robot or controller can learn a policy by receiving rewards for desirable states and actions.

M3.02 — Q-learning (2 hours)

Concept and explanation

Q-learning estimates the long-term value of taking an action in a state. The Q-value is updated from the immediate reward and the best estimated value of the next state, subject to a learning rate and discount factor.

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Study points

- State the meaning of a Q-value.
- Explain the update idea.
- Relate a policy to the largest Q-value.

Illustrative example: A table of state-action values can be updated after each transition until the action choices become stable.

M3.03 — Regression evaluation metrics (2 hours)

Concept and explanation

Mean Absolute Error averages absolute deviations, Mean Squared Error squares them, Root Mean Squared Error returns the error to the target unit, and R-squared describes explained variation relative to a baseline.

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Study points

- Define MAE, MSE, RMSE and R-squared.
- State the unit of each error measure.
- Choose a metric for a stated cost.

> **Illustrative example:** If errors are measured in kilopascals, MAE and RMSE are also expressed in kilopascals, while MSE has squared units.

M3.04 — Classification evaluation metrics (2 hours)

Concept and explanation

A confusion matrix counts true positives, true negatives, false positives and false negatives. Accuracy, precision, recall, F-score and the ROC curve summarize different aspects of classification performance.

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Study points

- Define precision and recall.
- Explain why accuracy may be misleading for imbalance.
- Interpret an ROC curve conceptually.

> **Common mistake:** Do not report accuracy alone when one class is rare or when false negatives have a high cost.

M3.05 — Clustering evaluation and validation (2 hours)

Concept and explanation

Sum of Squared Error measures within-cluster dispersion, while the Silhouette Coefficient and Dunn’s Index compare cohesion and separation. Ensemble methods, bootstrapping and cross-validation help assess stability or generalisation.

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Study points

- Compare cohesion and separation.
- State the purpose of bootstrapping.
- Explain why cross-validation must respect the data structure.

> **Module summary:** A metric is meaningful only when its definition, data split and decision cost are understood together.

OFFICIAL MODULE 4

Artificial Neural Networks

An artificial neural network represents a function through connected processing units. The module progresses from representation and perceptrons to multilayer networks, back propagation, recurrent networks, loss, activation and batch normalization.

Hours: 16
CO4 · Applying · Bloom level: Applying

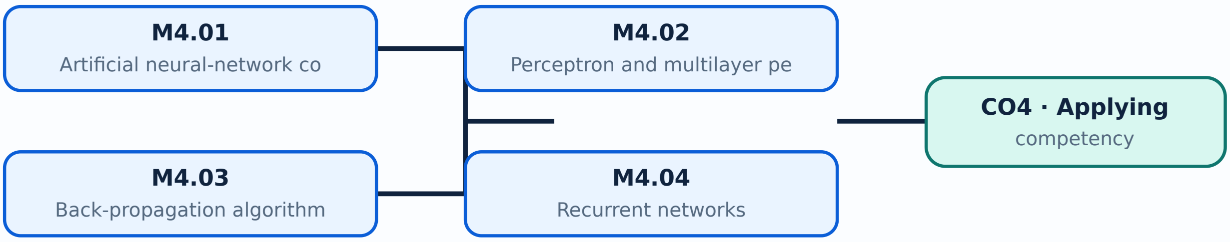
Learning goals

- Describe the structure of an ANN.
- Explain perceptron learning and back propagation.

- Relate activation and loss functions to training.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

M4.01 — Artificial neural-network concepts (3 hours)

Concept and explanation

A neural network contains input units, hidden layers and output units connected by weighted links. A weighted sum followed by an activation function produces the next layer’s output.

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Study points

- Identify layers and weights.
- Explain forward propagation.
- State one advantage and limitation of ANNs.

Engineering / workplace application: Image, speech and sensor-signal classification commonly use neural networks.

M4.02 — Perceptron and multilayer perceptron (3 hours)

Concept and explanation

A perceptron is a basic linear decision unit whose weights are adjusted from prediction error. A multilayer perceptron uses one or more hidden layers and nonlinear activations to represent more complex relationships.

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Study points

- Explain perceptron training.
- Distinguish a single layer from a multilayer network.
- Relate nonlinearity to model capacity.

M4.03 — Back-propagation algorithm (3 hours)

Concept and explanation

Back propagation applies the chain rule to calculate how the loss changes with respect to each weight. An optimizer then updates the weights in a direction that reduces the loss.

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Study points

- Describe forward and backward passes.
- Explain the role of a learning rate.
- Identify overfitting as a training risk.

Suggested procedure

1. Compute the forward output.
2. Evaluate the loss.
3. Propagate gradients backward.
4. Update weights and repeat for batches or epochs.

M4.04 — Recurrent networks (3 hours)

Concept and explanation

Recurrent neural networks retain a hidden state so that earlier inputs can influence later outputs. This makes them suitable for ordered sequences, although long sequences may create vanishing or exploding gradient issues.

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Study points

- Explain a recurrent state.
- Give a sequence-processing application.
- State one training difficulty.

M4.05 — Loss and activation functions (2 hours)

Concept and explanation

A loss function measures disagreement between the prediction and target. Activation functions such as sigmoid, tanh and ReLU introduce nonlinear behaviour and influence gradient flow and output interpretation.

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Study points

- Relate loss to the learning objective.
- Choose an output activation conceptually.
- Distinguish activation from loss.

M4.06 — Batch normalization (2 hours)

Concept and explanation

Batch normalization normalizes intermediate activations during training and uses learnable scale and shift parameters. It can improve optimization stability, but training and inference modes must be handled correctly.

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Study points

- State the purpose of batch normalization.
- Distinguish training and inference behaviour.
- Mention one implementation caution.

Module summary: Neural-network performance depends on architecture, data, initialization, optimization, regularization and a suitable evaluation protocol.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

...s labelled learning-map diagram.

Module 2: Clustering and

...rning-map diagram.

Module 3: Evaluation

...module to view its labelled learning-map diagram.

Module 4: Artificial Neural

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- No separate activity list is provided in the supplied PDF.

Practical requirements / equipment

- No separate practical equipment list is provided in the supplied PDF.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Series Test: 2 hours/assessment component as stated in the supplied syllabus.
- The supplied syllabus does not state a separate CIE/ESE mark distribution.

Module-wise Questions and Answers

module-1 — Supervised machine learning techniques

1. Explain Machine learning concepts. **[3 marks]**

Machine learning is the study of algorithms that learn patterns from examples rather than being programmed with every rule. Applications include prediction, classification, recommendation, anomaly detection and automation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Process involved in machine learning. **[3 marks]**

A typical process includes defining the problem, collecting and understanding data, cleaning and preprocessing, selecting features, dividing data into training and testing sets, training a model, evaluating it and monitoring performance.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Supervised learning. [3 marks]

Supervised learning uses labelled examples containing input features and a known target. Classification predicts categories, whereas regression predicts a numerical value.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Classification techniques. [3 marks]

Classification methods include logistic regression, K-nearest neighbours, support vector machines, Naive Bayes, decision trees and random forests. Model choice depends on data type, interpretability, scale, noise and performance requirements.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Clustering and dimensionality reduction

5. Explain Types of clustering. [3 marks]

Clustering groups observations so that members of a group are more similar to one another than to members of other groups. Hierarchical, agglomerative, divisive and partitional approaches differ in how groups are formed and represented.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Clustering techniques. [3 marks]

K-means iteratively assigns points to the nearest centroid and updates centroids. Mean-shift moves candidate centres toward regions of high density. Initialisation, distance measure, scaling and the number of clusters affect the result.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Dimensionality reduction techniques. [3 marks]

Principal Component Analysis creates orthogonal directions of maximum variance. Factor analysis models observed variables through latent factors, multidimensional scaling preserves pairwise distances, and Linear Discriminant Analysis seeks class-separating directions when labels are available.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Evaluation metrics for machine-learning models

8. Explain Reinforcement learning. [3 marks]

Reinforcement learning involves an agent, environment, state, action, reward and policy. The agent learns through interaction rather than a fixed labelled dataset; the objective balances exploration and exploitation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

9. Explain Q-learning. [3 marks]

Q-learning estimates the long-term value of taking an action in a state. The Q-value is updated from the immediate reward and the best estimated value of the next state, subject to a learning rate and discount factor.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Regression evaluation metrics. **[3 marks]**

Mean Absolute Error averages absolute deviations, Mean Squared Error squares them, Root Mean Squared Error returns the error to the target unit, and R-squared describes explained variation relative to a baseline.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Classification evaluation metrics. **[3 marks]**

A confusion matrix counts true positives, true negatives, false positives and false negatives. Accuracy, precision, recall, F-score and the ROC curve summarize different aspects of classification performance.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Artificial Neural Networks

12. Explain Artificial neural-network concepts. **[3 marks]**

A neural network contains input units, hidden layers and output units connected by weighted links. A weighted sum followed by an activation function produces the next layer’s output.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

13. Explain Perceptron and multilayer perceptron. **[3 marks]**

A perceptron is a basic linear decision unit whose weights are adjusted from prediction error. A multilayer perceptron uses one or more hidden layers and nonlinear activations to represent more complex relationships.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Back-propagation algorithm. **[3 marks]**

Back propagation applies the chain rule to calculate how the loss changes with respect to each weight. An optimizer then updates the weights in a direction that reduces the loss.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Recurrent networks. **[3 marks]**

Recurrent neural networks retain a hidden state so that earlier inputs can influence later outputs. This makes them suitable for ordered sequences, although long sequences may create vanishing or exploding gradient issues.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Machine learning concepts* with one relevant application. (5 marks)
2. **2.** Define or explain *Process involved in machine learning* with one relevant application. (5 marks)
3. **3.** Define or explain *Types of clustering* with one relevant application. (5 marks)
4. **4.** Define or explain *Clustering techniques* with one relevant application. (5 marks)
5. **5.** Define or explain *Reinforcement learning* with one relevant application. (5 marks)
6. **6.** Define or explain *Q-learning* with one relevant application. (5 marks)
7. **7.** Define or explain *Artificial neural-network concepts* with one relevant application. (5 marks)
8. **8.** Define or explain *Perceptron and multilayer perceptron* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Supervised machine learning techniques:** A reliable workflow moves from problem definition and data preparation to training, validation, evaluation and deployment.
- **Clustering and dimensionality reduction:** Unsupervised analysis is exploratory: the result must be interpreted against the data and engineering context.
- **Evaluation metrics for machine-learning models:** A metric is meaningful only when its definition, data split and decision cost are understood together.
- **Artificial Neural Networks:** Neural-network performance depends on architecture, data, initialization, optimization, regularization and a suitable evaluation protocol.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Machine learning concepts	Machine learning is the study of algorithms that learn patterns from examples rather than being programmed with every rule. Applications include prediction, classification,

Term	Short meaning
	recommendation, anomaly detection and automation.
Process involved in machine learning	A typical process includes defining the problem, collecting and understanding data, cleaning and preprocessing, selecting features, dividing data into training and testing sets, training a model, evaluating it and monitoring performance.
Supervised learning	Supervised learning uses labelled examples containing input features and a known target. Classification predicts categories, whereas regression predicts a numerical value.
Classification techniques	Classification methods include logistic regression, K-nearest neighbours, support vector machines, Naive Bayes, decision trees and random forests. Model choice depends on data type, interpretability, scale, noise and performance requirements.
Regression techniques	Linear regression models a numerical response using one or more input variables. Multiple linear regression extends the model to several predictors and requires attention to scale, correlation, residuals and overfitting.
Data preprocessing	Preprocessing converts raw data into a form suitable for learning. Normalization brings variables to comparable ranges, while missing-value handling may use deletion, imputation or a domain-informed replacement.
Types of clustering	Clustering groups observations so that members of a group are more similar to one another than to members of other groups. Hierarchical, agglomerative, divisive and partitional approaches differ in how groups are formed and represented.
Clustering techniques	K-means iteratively assigns points to the nearest centroid and updates centroids. Mean-shift moves candidate centres toward regions of high density. Initialisation, distance measure, scaling and the number of clusters affect the result.
Dimensionality reduction techniques	Principal Component Analysis creates orthogonal directions of maximum variance. Factor analysis models observed variables through latent factors, multidimensional scaling preserves pairwise distances, and Linear Discriminant Analysis seeks class-separating dir
Reinforcement learning	Reinforcement learning involves an agent, environment, state, action, reward and policy. The agent learns through interaction rather than a fixed labelled dataset; the objective balances exploration and exploitation.
Q-learning	Q-learning estimates the long-term value of taking an action in a state. The Q-value is updated from the immediate reward and the best estimated value of the next state, subject to a learning rate and discount factor.
Regression evaluation metrics	Mean Absolute Error averages absolute deviations, Mean Squared Error squares them, Root Mean Squared Error returns the error to the target unit, and R-squared describes explained variation relative to a baseline.
Classification evaluation metrics	A confusion matrix counts true positives, true negatives, false positives and false negatives. Accuracy, precision, recall, F-score and the ROC curve summarize different aspects of classification performance.
Clustering evaluation and validation	Sum of Squared Error measures within-cluster dispersion, while the Silhouette Coefficient and Dunn’s Index compare cohesion and separation. Ensemble methods, bootstrapping and cross-validation help assess stability or generalisation.

Term	Short meaning
Artificial neural-network concepts	A neural network contains input units, hidden layers and output units connected by weighted links. A weighted sum followed by an activation function produces the next layer’s output.
Perceptron and multilayer perceptron	A perceptron is a basic linear decision unit whose weights are adjusted from prediction error. A multilayer perceptron uses one or more hidden layers and nonlinear activations to represent more complex relationships.
Back-propagation algorithm	Back propagation applies the chain rule to calculate how the loss changes with respect to each weight. An optimizer then updates the weights in a direction that reduces the loss.
Recurrent networks	Recurrent neural networks retain a hidden state so that earlier inputs can influence later outputs. This makes them suitable for ordered sequences, although long sequences may create vanishing or exploding gradient issues.
Loss and activation functions	A loss function measures disagreement between the prediction and target. Activation functions such as sigmoid, tanh and ReLU introduce nonlinear behaviour and influence gradient flow and output interpretation.
Batch normalization	Batch normalization normalizes intermediate activations during training and uses learnable scale and shift parameters. It can improve optimization stability, but training and inference modes must be handled correctly.

Official References and Resources

References listed in the supplied syllabus

1. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012.
2. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press, 2010.
3. Mitchell M. T., Machine Learning, McGraw Hill, 1997.
4. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell J., Machine Learning, Neural and Statistical Classification, Overseas Press, 1994.

Official learning resources listed

- https://www.javatpoint.com/machine-learning-models
- Introduction to Machine Learning - Course (nptel.ac.in)

In-page Search